
Machine Learning Fundamentals (Theory)

Welcome to **Machine Learning Basics**! In this session, You will learn what machine learning is, why we need it, how to set up your Python data science environment, the structure of an ML pipeline, and the core distinction between Regression and Classification.

<https://docs.google.com/presentation/d/1-LfpcP22p579HPtueV0Fxl-UnEOnepylKC0foT-Bm9U/edit?usp=sharing>

1. Machine Learning Environment Setup

Before training models, ensure your Python environment is equipped with essential data science libraries (`pandas` , `numpy` , `scikit-learn` , `joblib`).

A. Environment Setup Steps

Step 1: Install Python

1. Download the latest stable version of Python from the [Official Python Website](#).
2. *Crucial*: During installation, check the box that says "**Add Python to PATH**".
3. Verify installation in your terminal:

```
python --version
```

Step 2: Install ML Packages

With your environment, Open your terminal and install the core data science libraries:

```
pip install pandas numpy scikit-learn joblib
```

Step 3: Install Visual Studio Code (VS Code)

VS Code is the preferred code editor for modern web development.

1. Download it from the [Visual Studio Code Website](#).
- *Essential Extensions*: Install the **Python** extension (by Microsoft) and **Pylance** for intelligent code completion, linting, and virtual environment management.
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2. What is Machine Learning & Why Do We Need It?

A. What is Machine Learning?

Traditional software development relies on explicit rule-based programming:

Data + Rules → Output

Machine Learning inverts this paradigm. Instead of writing rules manually, you feed historical data and answers into an algorithm so it can *discover* the rules:

Data + Answers → Rules (Model)

B. Why Do We Need Machine Learning?

- **Handling Complexity:** Some problems (like predicting house prices based on dozens of fluctuating factors, or classifying spam emails) are too complex to write manual `if-else` rules for.
- **Adaptability:** ML models can adapt and retrain on new data over time, improving accuracy automatically.
- **Automation & Prediction:** Enables software to forecast future outcomes (e.g., customer churn, sales volume) proactively.

C. Types of Machine Learning

1. **Supervised Learning:** The model is trained on labeled data (input features + correct target answers). *This is what we will focus on for our mini-projects.*
2. **Unsupervised Learning:** The model finds hidden patterns or groupings in unlabeled data (e.g., customer segmentation via clustering).
3. **Reinforcement Learning:** An agent learns to make decisions by performing actions and receiving rewards or penalties.

3. Overview of the Machine Learning Pipeline

An ML pipeline is the systematic workflow used to take raw data and turn it into a deployed, predictive model. It consists of three primary stages:

1. **Data Preprocessing:** Raw data is rarely ready for algorithms straight out of the box. Key data preprocessing steps include:
 - **Data Cleaning & Imputation:** Handling missing data (often represented as `NaN`) without losing valuable rows. An **Imputer** (like scikit-learn's `SimpleImputer`) is used to mathematically fill these gaps.
 - *Mean / Median Imputation:* Replaces missing numerical data with the column average (mean) or middle value (median).

- *Mode Imputation*: Replaces missing categorical data with the most frequent value.
- **Encoding Categorical Data**: Machine learning models only understand numbers, so text labels must be converted.
 - *Ordinal Encoding*: Assigns a unique integer to categories (e.g., "Low"=0, "Medium"=1, "High"=2). Best for ordinal data with a clear hierarchy.
 - *One-Hot Encoding*: Creates separate binary (0/1) columns for each category so the model doesn't assume a false numerical order (e.g., turning a "Color" column into "Is_Red", "Is_Blue").
- **Feature Scaling**: Normalizing numerical variables so features with large numbers (e.g., house square footage: 2000) don't overpower features with smaller numbers (e.g., number of rooms: 3).
 - *Standardization (StandardScaler)*: Shifts data so it has a mean of 0 and a standard deviation of 1.
 - *Normalization (MinMaxScaler)*: Squeezes all numerical data into a strict range, usually between 0 and 1.

2. Feature Selection:

- Identifying and selecting only the most relevant columns/features that contribute strongly to predicting the target variable, reducing noise and training time.

3. Train/Test Split:

- Splitting your dataset into two parts: a **Training Set** (usually 80% of the data) used to teach the model, and a **Testing Set** (20%) kept hidden to evaluate how well the model generalizes to unseen data.

4. Regression vs. Classification

Supervised learning algorithms are broadly divided into two main categories based on what type of target variable they predict:

A. Regression (Predicting Continuous Values)

- **Goal**: Predict a continuous, numerical quantity.
- **Examples**: Predicting house prices, employee salaries, car fuel efficiency (MPG), or exam scores.
- **Common Algorithms**: Linear Regression, Ridge/Lasso Regression, Decision Tree Regressor.

B. Classification (Predicting Categories)

- **Goal:** Predict a discrete class label or category.
 - **Examples:** Predicting whether an email is `Spam` or `Not Spam`, whether a customer will `Churn` (`Yes` or `No`), or diagnosing disease risk (`High Risk` vs. `Low Risk` vs. `No Risk`).
 - **Common Algorithms:** Logistic Regression, Decision Tree Classifier, Random Forest.
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5. Reference Links & Further Reading

- **Python Getting Started Guide:** Official documentation for Python installation and virtual environments.
 - **Scikit-Learn Getting Started Guide:** Official tutorial for machine learning in Python.
 - **Pandas Documentation:** Essential guide for data manipulation and preprocessing.
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